Evaluation of Ontology-Grounded Nutrition Recommendations

The purpose of this evaluation is to assess the quality of nutrition recommendations generated from the system which are based on the SustainHealthyFood Ontology. For each persona and system response, please evaluate how well the answer complies with the user profile and nutrition requirements. The main purpose is to have fully personalized recommendations.

To get an overview of what the system looks like and how it works, please check the following video: https://youtu.be/cugVFGgGUlk.

About the system:

The SustainHealthyFood Ontology is operationalized as a knowledge graph: its food, recipe, and nutritional information is populated with data (triples) sourced from the NEVO dataset (Dutch Food Composition Database). Because food and recipe names are carried over from NEVO's own naming conventions rather than rewritten, you'll often see abbreviations in suggestions — e.g. "raw" / "prepared" for preparation state, or "w" / "wo" for "with" / "without" an ingredient.

Nutritional data source:

Rijksinstituut voor Volksgezondheid en Milieu (RIVM). (2023). *NEVO online, version 2023/8.0*. RIVM, Bilthoven.

https://www.rivm.nl/en/dutch-food-composition-database

* Indicates required question

Consent for participation

About this study

This study is conducted by Donika Xhani, PhD candidate at the University of Twente, and evaluates the quality of answers generated by an ontology-grounded Retrieval-Augmented Generation (RAG) system.

You will be asked to rate a set of system-generated answers and provide an overall evaluation of the system. The questionnaire takes approximately 25 **minutes**.

Participation is voluntary. You are not being tested; the **system output is being evaluated**. You may stop at any time before submitting the form.

The study does not intentionally collect your name, email address, or other direct identifying information. Your ratings and optional comments will be used for research purposes, including a research paper, presentations, and a PhD thesis. Data will be stored securely at the University of Twente for **5 years** and reported only in aggregate or de-identified form.

Please do not include identifying information in free-text comments. Because responses are collected without direct identifiers, it may not be possible to identify and remove your individual response after submission.

There is no compensation for participation, and no significant risks are anticipated.

1. Consent Statement *

Please tick the box below to confirm that you read and understand this study.

Tick all that apply.

☐ I have read and understood the information above, and I voluntarily consent to participate in this study

Expert Background Information

2. What is your primary professional background?

Mark only one oval.

- Registered dietitian / Nutritionist
- Nutrition researcher
- Food scientist/ specialist
- Other: _____

3. How many years of professional experience do you have in nutrition, dietetics, or a related field?

Mark only one oval.

- Less than 2 years
- 2–5 years
- 6–10 years
- More than 10 years

4. Do you currently use any digital applications, software systems, or AI-based tools (e.g., chatbots or recommender systems) as part of your work?

Mark only one oval.

- Yes
- No
- I have used them in the past, but not currently

Skip to question 5

Section 1

5. A 30-year old female who is Pregnant and has a Dairy Intolerance wants to get information which food are recommended or not recommended for pregnancy. She asks the system the following question: * "*Which foods are recommended and not recommended for pregnancy ?*"

   This is the **generated answer from the system**:
   Based on the SustainHealthyFood Ontology for Pregnancy, here are the recommended and not recommended foods:

   RECOMMENDED for Pregnancy:

   -Beans French boiled: Rich in folate, which helps prevent birth defects.

   -Beef: High-quality protein that supports fetal growth and development.

   -Broccoli boiled: Packed with vitamin C, which boosts immune function during pregnancy.

   -Chicken: Good source of iron, essential for healthy red blood cells.

   -Chicken fillet prepared: Lean protein that supports muscle growth in the fetus.

   -Egg whole chicken av boiled: Rich in choline, crucial for fetal brain development.

   -Lentils green and brown boiled: High-quality plant-based protein that supports overall health.

   -Salad salmon w potato: Rich in omega-3 fatty acids, which support fetal brain and eye development.

   -Salmon smoked: Packed with vitamin D, essential for bone growth and development.

   -Spinach boiled: Rich in iron, supporting healthy red blood cells.

   NOT RECOMMENDED / AVOID for Pregnancy:

   -Beer pilsner

   -Brandy

   -Liqueur 15-25 vol% alcohol

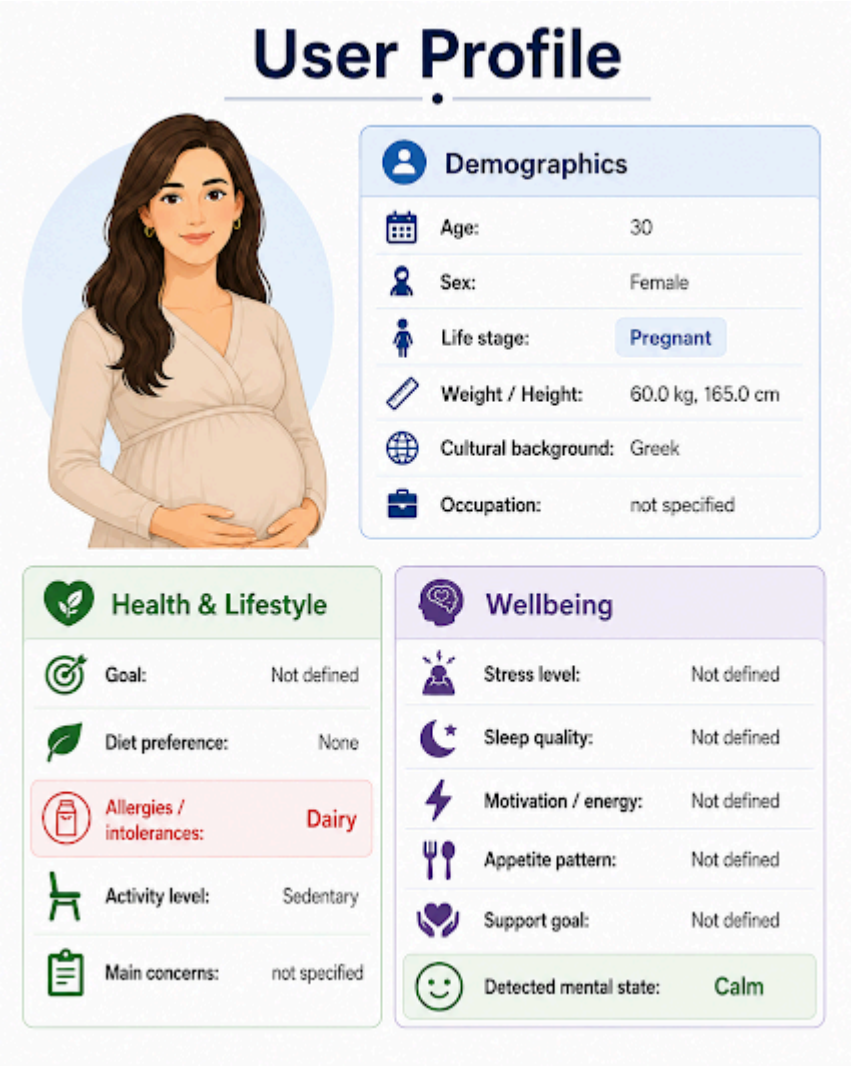
   -Whisky

   -Wine white dry

   -Wine white sweet

These recommendations are based on the provided ontology context and your stated profile, but may not account for your complete medical history or individual circumstances. Always consult a healthcare professional for specific dietary advice.

Based on the attached full user profile and the generated answer, evaluate the following statements:



Mark only one oval per row.

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
The answer is personalized to the user's profile	◯	◯	◯	◯	◯
The recommended food are nutritionally appropriate.	◯	◯	◯	◯	◯
The answer avoids inappropriate or	◯	◯	◯	◯	◯

unsafe recommendations.

6. Please briefly explain the reasoning behind your ratings, particularly if you identified any issues with the recommendations.

Section 2

7. A 30-year old female who has a Dairy Intolerance is worried about her iron deficiency. She asks the system the following question: * "What foods are recommended for someone with iron deficiency(anemia)?"

This is the **generated answer from the system**:
For someone with iron deficiency, the following foods are recommended:

HIGHLY RECOMMENDED for Iron Deficiency:

-Beef >10% fat prepared av: Rich in heme iron, a type of iron that is easily absorbed by the body.

-Beef T-bone steak prepared: High in heme iron and other essential nutrients like zinc and vitamin B12.

-Beef T-bone steak raw: Contains heme iron and other beneficial compounds like conjugated linoleic acid (CLA).

-Beef av raw: Rich in non-heme iron, which is still easily absorbed with adequate vitamin C.

-Beef fat: Good source of heme iron and other essential fatty acids.

-Beef frying steak prepared: High in heme iron and other nutrients like zinc and selenium.

-Beef frying steak raw: Contains heme iron and beneficial compounds like CLA.

-Beef olives prepared: Rich in non-heme iron and other essential minerals like potassium.

-Beef olives raw: Good source of non-heme iron and antioxidants.

-Beef prime rib prepared: High in heme iron and other nutrients like zinc and vitamin B12.

-Beef prime rib raw: Contains heme iron and beneficial compounds like CLA.

-Beef rib prepared: Rich in heme iron and other essential minerals like selenium.

-Beef rib raw: Good source of non-heme iron and antioxidants.

-Beef rib steak prepared: High in heme iron and other nutrients like zinc and selenium.

-Beef rib steak raw: Contains heme iron and beneficial compounds like CLA.

-Beef ribeye steak prepared: Rich in heme iron and other essential minerals like potassium.

-Beef ribeye steak raw: Good source of non-heme iron and antioxidants.

-Beef roast prepared: High in heme iron and other nutrients like zinc and vitamin B12.

-Beef roast raw: Contains heme iron and beneficial compounds like CLA.

-Beef rolled prepared: Rich in non-heme iron and other essential minerals like selenium.

-Beef rump steak prepared: Good source of heme iron and other beneficial compounds like CLA.

-Beef rump steak raw: Contains non-heme iron and antioxidants.

-Beef salted cooked: High in heme iron and other nutrients like zinc and selenium.

-Beef salted cooked Surinam: Rich in non-heme iron and other essential minerals like potassium.

-Beef shank raw: Good source of non-heme iron and antioxidants.

-Beef sirloin rolled raw: Contains heme iron and beneficial compounds like CLA.

-Beef smoke-dried: Rich in non-heme iron and other essential minerals like selenium.

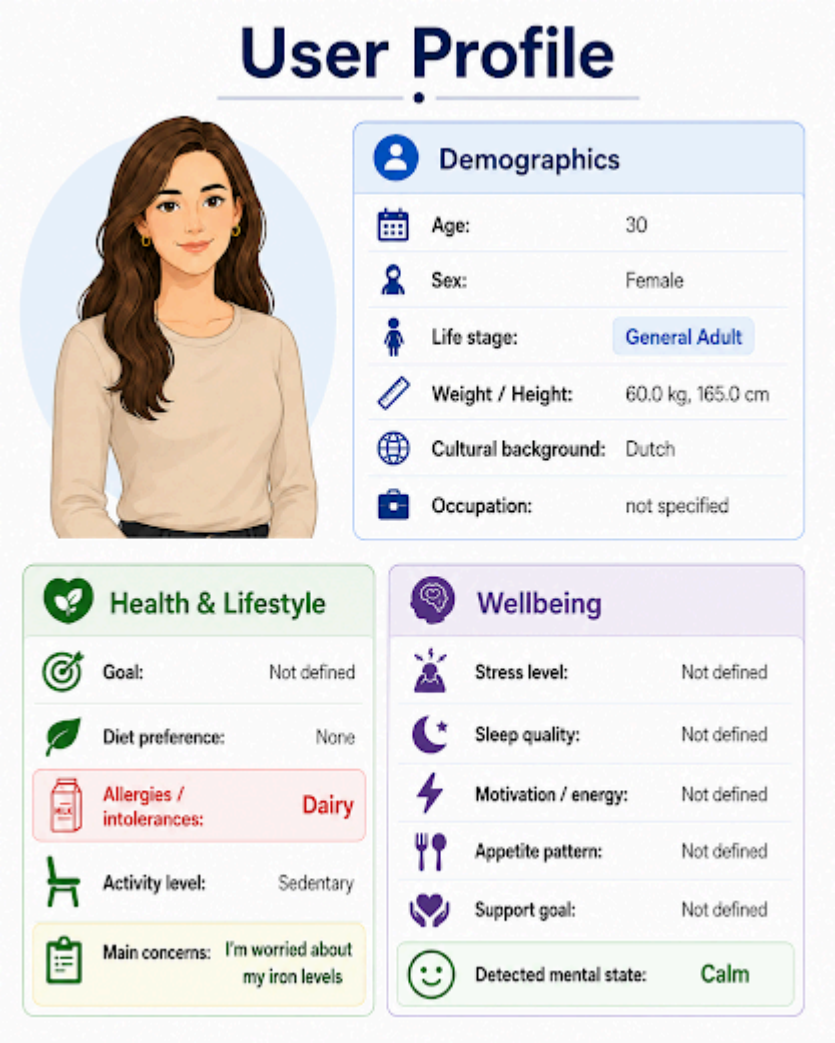
-Beef steak tartare prepared: High in heme iron and other nutrients like zinc and vitamin B12.

-Beef steak tartare raw: Contains heme iron and beneficial compounds like CLA.

-Beef steak tartare spiced filet americain: Rich in non-heme iron and other essential minerals like potassium.

These recommendations are based on the provided ontology context and your stated profile, but may not account for your complete medical history or individual circumstances. Always consult a healthcare professional for specific dietary advice.

Based on the attached full user profile and the generated answer, evaluate the following statements:



Mark only one oval per row.

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
The answer is personalized to the user's profile	◯	◯	◯	◯	◯
The recommended food are nutritionally appropriate.	◯	◯	◯	◯	◯
The answer avoids inappropriate or unsafe recommendations.	◯	◯	◯	◯	◯

8. Please briefly explain the reasoning behind your ratings, particularly if you identified any issues with the recommendations.

Section 3

9. A 30-year old female who is Vegan, is worried about her iron deficiency. She asks the system the following question: * "What foods are recommended for someone with iron deficiency(anemia)?"

This is the **generated answer from the system**:
For someone with iron deficiency, HIGHLY RECOMMENDED foods are:

-Lentils green and brown boiled: Rich in iron, which helps increase red blood cell production.

-Lentils green and brown dried: Similar to boiled lentils, providing a good source of iron.

-Lentils red boiled: Another type of lentil that is high in iron, supporting healthy red blood cells.

-Pumpkin seeds: A good source of non-heme iron, which the body can use more easily.

-Spinach boiled: Packed with iron and other essential nutrients for overall health.

-Spinach creamed frozen boiled: Similar to fresh spinach, providing a boost of iron and vitamins.

-Spinach frozen boiled: Convenient and nutritious, making it an excellent addition to meals.

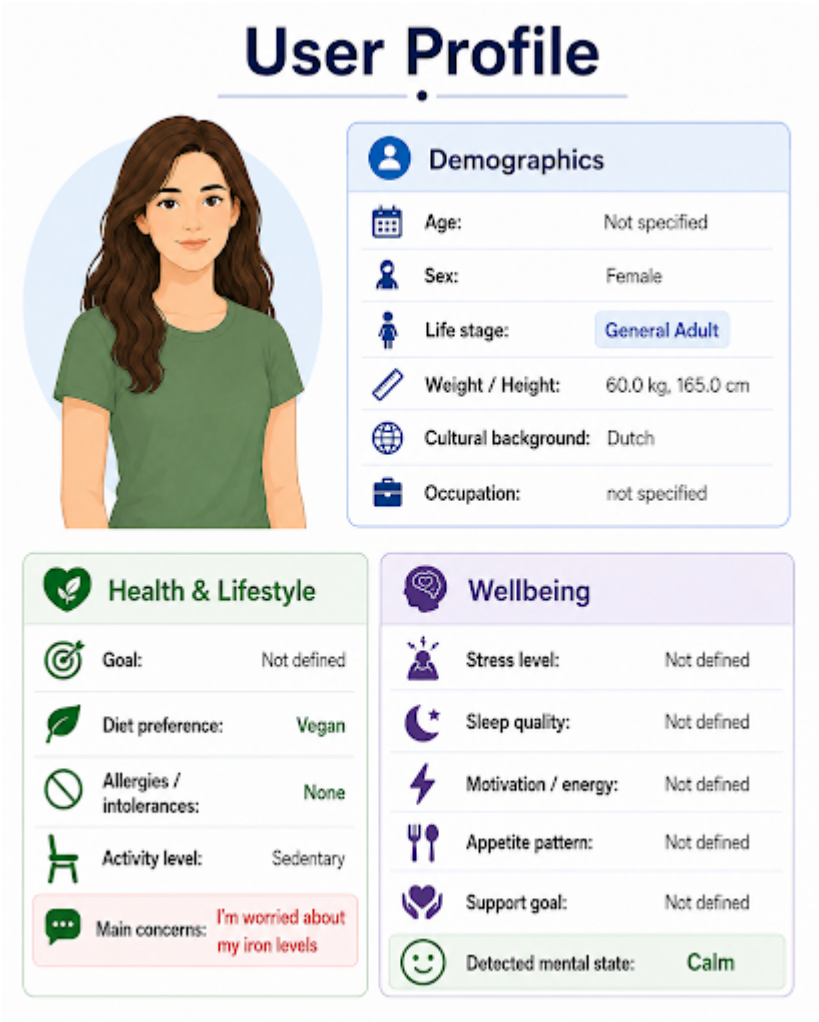
-Spinach tinned: Still rich in iron and other nutrients despite being canned.

Some foods were excluded from these recommendations due to conflicting nutritional profiles with the user's Vegan profile.

ℹ️ **Note:** 67 food(s) that would otherwise appear above were excluded because they conflict with your Vegan profile.

These recommendations are based on the provided ontology context and your stated profile, but may not account for your complete medical history or individual circumstances. Always consult a healthcare professional for specific dietary advice.

Based on the attached full user profile and the generated answer, evaluate the following statements:



Mark only one oval per row.

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
The answer is personalized to the user's profile	◯	◯	◯	◯	◯
The recommended food are nutritionally appropriate.	◯	◯	◯	◯	◯
The answer avoids inappropriate or unsafe recommendations.	◯	◯	◯	◯	◯

10. Please briefly explain the reasoning behind your ratings, particularly if you identified any issues with the recommendations.

Section 4

11. A dutch male who is Vegan and has type 2 diabetes. He asks the system the following question: *
"Which foods should I limit and which should I eat more of if I have type 2 diabetes?"
This is the **generated answer from the system**:
Based on the SustainHealthyFood Ontology for Diabetes, here are the recommended and not recommended foods:

Recommended for Diabetes:

-Apple Elstar w skin helps by providing fiber and antioxidants.

-Apple Jonagold w skin helps by providing fiber and antioxidants.

-Apple sauce wo sugar tinned helps by reducing added sugars.

-Asparagus white boiled helps by providing vitamin C and potassium.

-Beans French boiled helps by providing protein, fiber, and minerals.

-Beans French frozen boiled helps by providing protein, fiber, and minerals.

-Bilberries helps by providing antioxidants and anthocyanins.

NOT RECOMMENDED / AVOID for Diabetes:

-Almond paste filled tarts w butter (conflicts with your profile)

-Almond paste filled tarts wo butter (conflicts with your profile)

-Almond paste w egg (conflicts with your profile)

-Appel pie Dutch w shortbread w butter (conflicts with your profile)

-Biscuit (conflicts with your profile)

-Biscuit Cafe noir (conflicts with your profile)

-Biscuit Dutch Amaretti Bitterkoekjes (conflicts with your profile)

-Biscuit Dutch Frou frou (conflicts with your profile)

-Biscuit Dutch Kletskop (conflicts with your profile)

-Biscuit Dutch Krakeling (conflicts with your profile)

-Biscuit Dutch shortbread sprits (conflicts with your profile)

-Biscuit Jaffa cakes/Cake PiM's (conflicts with your profile)

-Biscuit av (conflicts with your profile)

-Biscuit brown/digestive w chocolate (conflicts with your profile)

-Biscuit brown/wholemeal (conflicts with your profile)

Key nutrients:

-Fiber helps regulate blood sugar levels and improve insulin sensitivity.

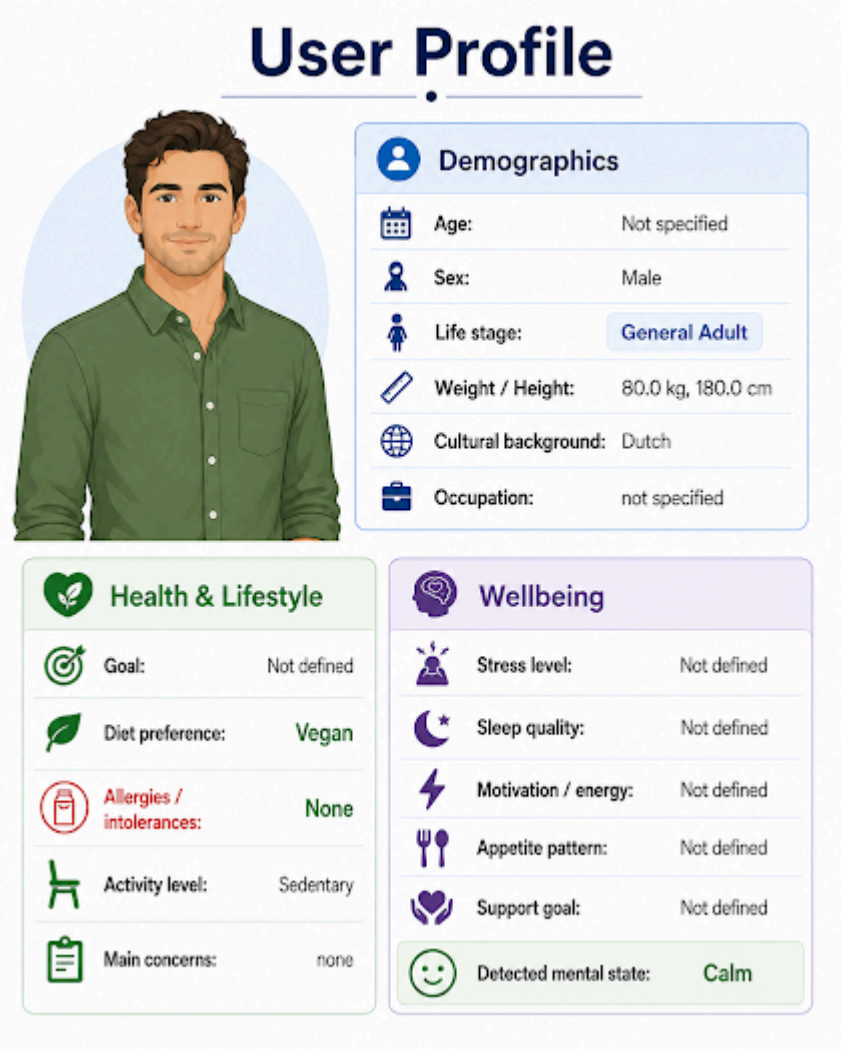
-Potassium helps lower blood pressure and reduce the risk of cardiovascular disease.

-Folate helps prevent homocysteine buildup, which can damage blood vessels.

ℹ️ **Note:** 73 food(s) that would otherwise appear above were excluded because they conflict with your Vegan profile.

These recommendations are based on the provided ontology context and your stated profile, but may not account for your complete medical history or individual circumstances. Always consult a healthcare professional for specific dietary advice.

Based on the attached full user profile and the generated answer, evaluate the following statements:



Mark only one oval per row.

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
The answer is personalized to the user's profile	◯	◯	◯	◯	◯
The recommended food are nutritionally appropriate.	◯	◯	◯	◯	◯
The answer avoids inappropriate or unsafe recommendations.	◯	◯	◯	◯	◯

12. Please briefly explain the reasoning behind your ratings, particularly if you identified any issues with the recommendations.

Section 5

13. A 40-year old dutch male has gluten allergy and type 2 diabetes. He asks the system the following question: *
"Which foods should I limit and which should I eat more of if I have type 2 diabetes?"
This is the **generated answer from the system**:
Based on the SustainHealthyFood Ontology for Diabetes, here are the recommended and not recommended foods:

Recommended for Diabetes:

-Apple Elstar w skin (high in fiber and antioxidants)

-Apple Jonagold w skin (rich in vitamins A and C)

-Beans French boiled (good source of protein and fiber)

-Beans kidney red boiled (high in potassium and fiber)

-Bilberries (antioxidant-rich and may help lower blood sugar)

-Asparagus white boiled (rich in vitamin C and fiber)

NOT RECOMMENDED / AVOID:

-Almond paste filled tarts w butter

-BiscuitBakery mix for Dutch little pancakes poffertjes

-Bakery mix for pancakes

-Bakery mix for pancakes wholemeal

-Biscuit Cafe noir

-Biscuit Dutch Amaretti Bitterkoekjes

-Biscuit Dutch Frou frou

-Biscuit Dutch Kletskop

-Biscuit Dutch Krakeling

-Biscuit Dutch shortbread sprits

-Biscuit Jaffa cakes/Cake PiM's

-Biscuit av

-Biscuit brown/digestive w chocolate

-Biscuit brown/wholemeal

Key nutrients:

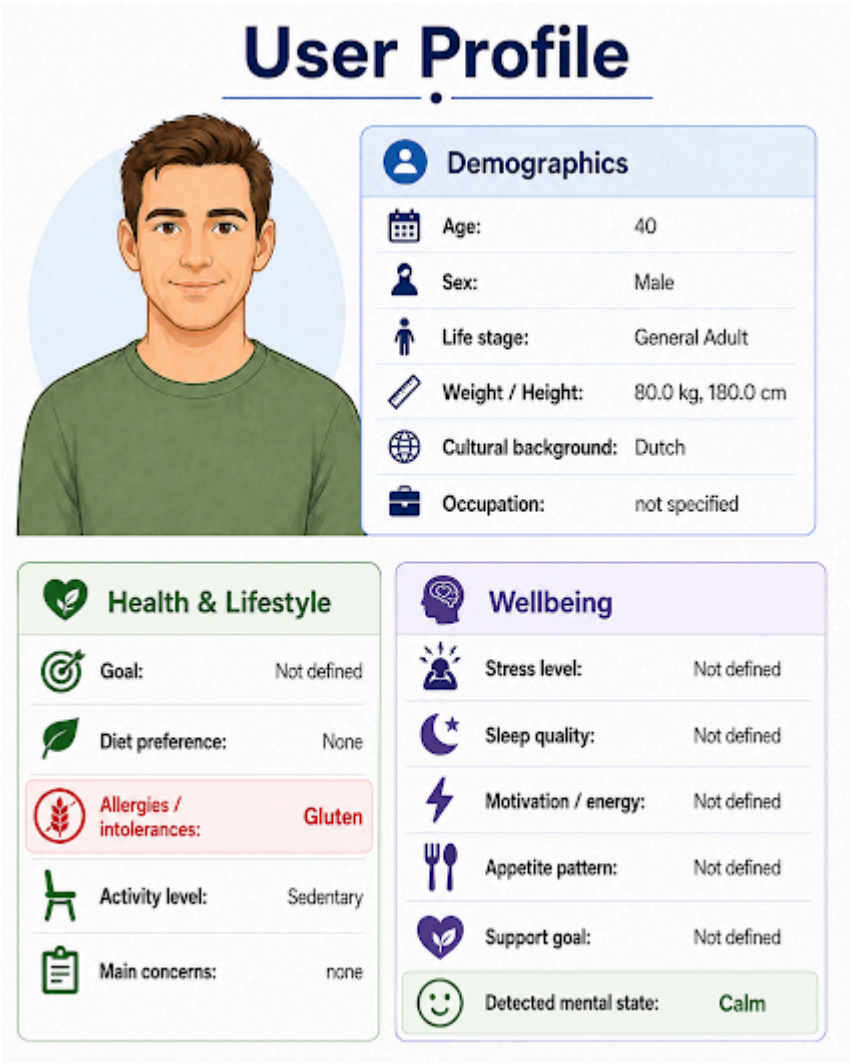
-Fiber (highly recommended for Diabetes)

-Potassium (highly recommended for Diabetes)

ℹ️ **Note:** 73 food(s) that would otherwise appear above were excluded because they conflict with your Vegan/Gluten profile.

These recommendations are based on the provided ontology context and your stated profile, but may not account for your complete medical history or individual circumstances. Always consult a healthcare professional for specific dietary advice.

Based on the attached full user profile and the generated answer, evaluate the following statements:



Mark only one oval per row.

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
The answer is					

The answer is personalized to the user's profile					
The recommended food are nutritionally appropriate.					
The answer avoids inappropriate or unsafe recommendations.					

14. Please briefly explain the reasoning behind your ratings, particularly if you identified any issues with the recommendations.

Section 6

15. A 30-year old dutch male who has gluten allergy is currenctly experiencing some stress. He asks the system the following question: * "What food or nutrition help to manage stress?" This is the **generated answer from the system**: Based on the SustainHealthyFood Ontology for Stressed, here are the recommended foods:

For Stressed:

HIGHLY RECOMMENDED for Stressed (47 total):

-Almonds blanched unsalted: Rich in magnesium, which helps regulate stress response.

-Beans kidney red dried: High in potassium, an essential mineral that supports healthy blood pressure and reduces stress.

-Brazil nuts salted: Good source of selenium, a nutrient that acts as an antioxidant to reduce oxidative stress.

-Cashew nuts unsalted: Contains magnesium and copper, both important for maintaining healthy nerve function under stress.

-Chia seeds dried: Rich in omega-3 fatty acids, which support brain health and may help alleviate stress symptoms.

-Cocoa powder: Contains flavonoids that have been shown to improve mood and reduce stress levels.

-Hazelnuts unsalted: Good source of magnesium and copper, supporting healthy nerve function under stress.

-Hemp seed: High in essential fatty acids, including omega-3 and omega-6, which support brain health and may help alleviate stress symptoms.

-Lentils green and brown dried: Rich in potassium, an essential mineral that supports healthy blood pressure and reduces stress.

-Linseeds: Good source of magnesium and copper, supporting healthy nerve function under stress.

-Oat flakes: High in fiber, which can help regulate blood sugar levels and reduce stress.

-Peanut butter: Contains magnesium and copper, both important for maintaining healthy nerve function under stress.

-Peanuts salted: Rich in potassium, an essential mineral that supports healthy blood pressure and reduces stress.

Key nutrients:

-Magnesium: Supports healthy nerve function and blood pressure regulation, reducing stress response.

-Potassium: Essential mineral that supports healthy blood pressure and reduces stress levels.

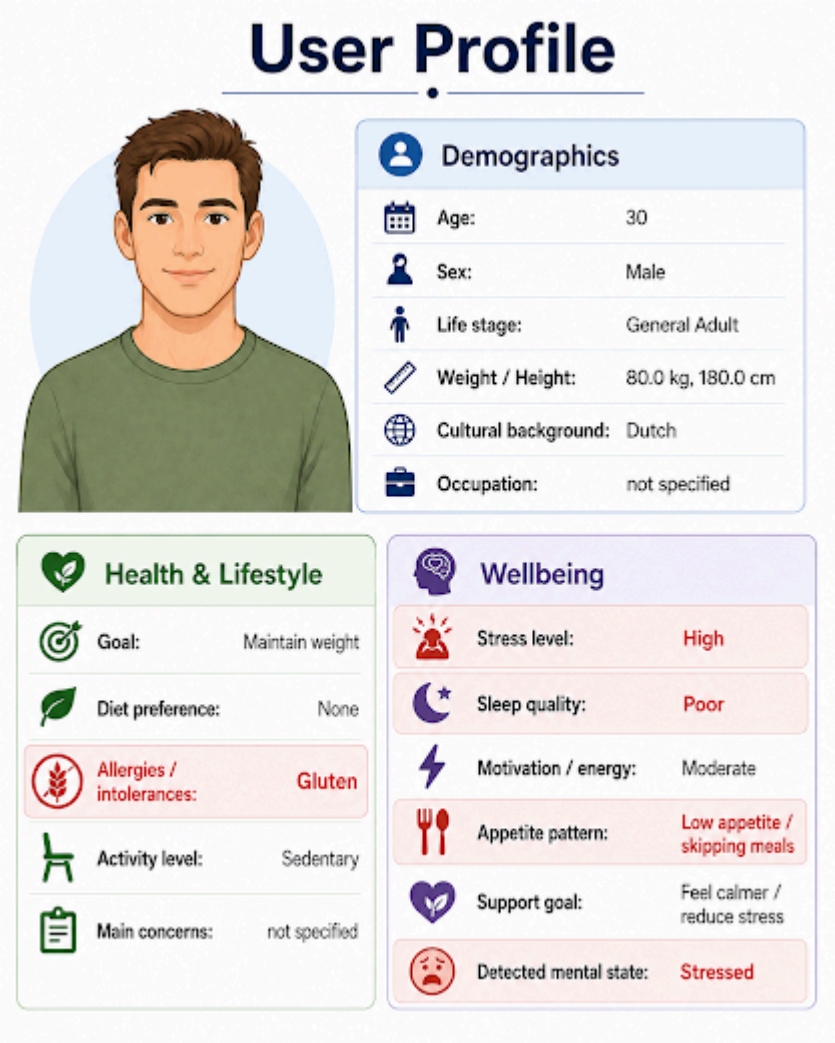
-Omega-3 fatty acids: Support brain health and may help alleviate stress symptoms.

-Copper: Important for maintaining healthy nerve function under stress.

ℹ️ **Note:** 22 food(s) that would otherwise appear above were excluded because they conflict with your Gluten profile.

These recommendations are based on the provided ontology context and your stated profile, but may not account for your complete medical history or individual circumstances. Always consult a healthcare professional for specific dietary advice.

Based on the attached full user profile and the generated answer, evaluate the following statements:



Mark only one oval per row.

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
The answer is personalized to the user's profile	◯	◯	◯	◯	◯
The recommended food are nutritionally appropriate.	◯	◯	◯	◯	◯
The answer avoids inappropriate or unsafe recommendations.	◯	◯	◯	◯	◯

16. Please briefly explain the reasoning behind your ratings, particularly if you identified any issues with the recommendations.

Section 7

17. In your opinion, what are the strengths and weaknesses of a system that gives these kinds of recommendations? *

18. Would it be useful to use such a system in your professional practise? Please elaborate on why or why not. *

19. Do you have any other comment about the recommendations or the system?

20. Thank you very much for doing this evaluation!

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